

《数据库系统原理》期末考试试题(A)

考试 注 意 事 项	一、学生参加考试须带学生证或学院证明，未带者不准进入考场。学生必须按照监考教师指定座位就坐。											
	二、书本、参考资料、书包等物品一律放到考场指定位置。											
三、学生不得另行携带、使用稿纸，要遵守《北京邮电大学考场规则》，有考场违纪或作弊行为者，按相应规定严肃处理。	四、学生必须将答题内容做在试题答卷上，做在试题及草稿纸上一律无效。											
	五、填空题用英文答，中文答对得一半分。											
考试 课程	数据库系统 原 理						考试时间		2016 年 6 月 21 日			
题号	一	二	三	四	五	六	七	八	九	十	十一	总分
满分	10	18	8	10	12	6	10	4	4	8	10	100
得分												
阅卷 教师												

1. Fill in blanks. (1×10 points)

(1) Nowadays, there are several commonly-used database system products. Among the following statements, the correct one/ones is/are _____.

A. SQL Server is Microsoft Corp's product, and Oracle and DB2 are developed by IBM.

B. MySQL belongs to open-source database systems.

C. SQL Server and Sybase are based on the relational model, Oracle and DB2 are based on the E-R model, and MySQL are based on the object-oriented model.

D. In a SQL Server database application system programmed in C, the application program can access SQL Server via application program interface, such as ODBC.

E. As a large scale database application system, 12306 Railway Ticket Booking System has a three-tier Browser-Server(B/S)-like architecture.

(2) The ability to modify the physical schema without changing the logical

schema or the views is _____ data Independence, that is application programs need not be rewritten if the physical schema changes.

(3) The foreign key values in any tuple of relation r_1 are either null or must appear as the _____ values of a tuple of the referenced relation r_2 .

(4) The six fundamental operations in the relational algebra are *select*, *project*, _____, *set difference*, *Cartesian product*, and *rename*.

(5) Given $R(U)$, $X, Y \subseteq U$, and the functional dependency $X \rightarrow Y$ holds on R . If there is a subset $X' \subset X$ and $X' \rightarrow Y$ holds on R , then $X \rightarrow Y$ is a _____ dependency.

(6) There are three possible ways of organizing records in files: heap file organization, _____, and hashing file organization.

(7) Query processing refers to the range of activities involved in extracting data from a database. The basic steps in processing a query are parsing and translation, _____, and evaluation.

(8) Transaction _____ property means that after a transaction completes successfully, the changes it has made to the database persist, even if there are system failures.

(9) The _____ two-phase locking protocol requires that all S-locks and X-locks are held until the transaction commits.

(10) _____, also called system catalog, stores metadata, e.g. data about data, such as names and types of attributes of each relation, number of tuples in each relation, names and definitions of views.

2. (18 points) The schema diagram in the database *University* is given below, and the primary keys in the relational tables are underlined. The tables *Department*, *Student*, *Teacher* and *Course* describe the four data objects in the database. A teacher in a department teaches a course in a class, and a student enrolls in a course, in the *Spring* or *Fall* Semester in a year. The table *Section* maintains the information on all classes (sections) taught. Each *Section* is identified by a *course_id*, *sec_id*, *year*, and *semester*, indicating that a course is taught in a class/section in the *Spring* or *Fall* semester in a year. The location and time of the course offering in a section are given by *building*, *room_no*, and *time_slot_id*.

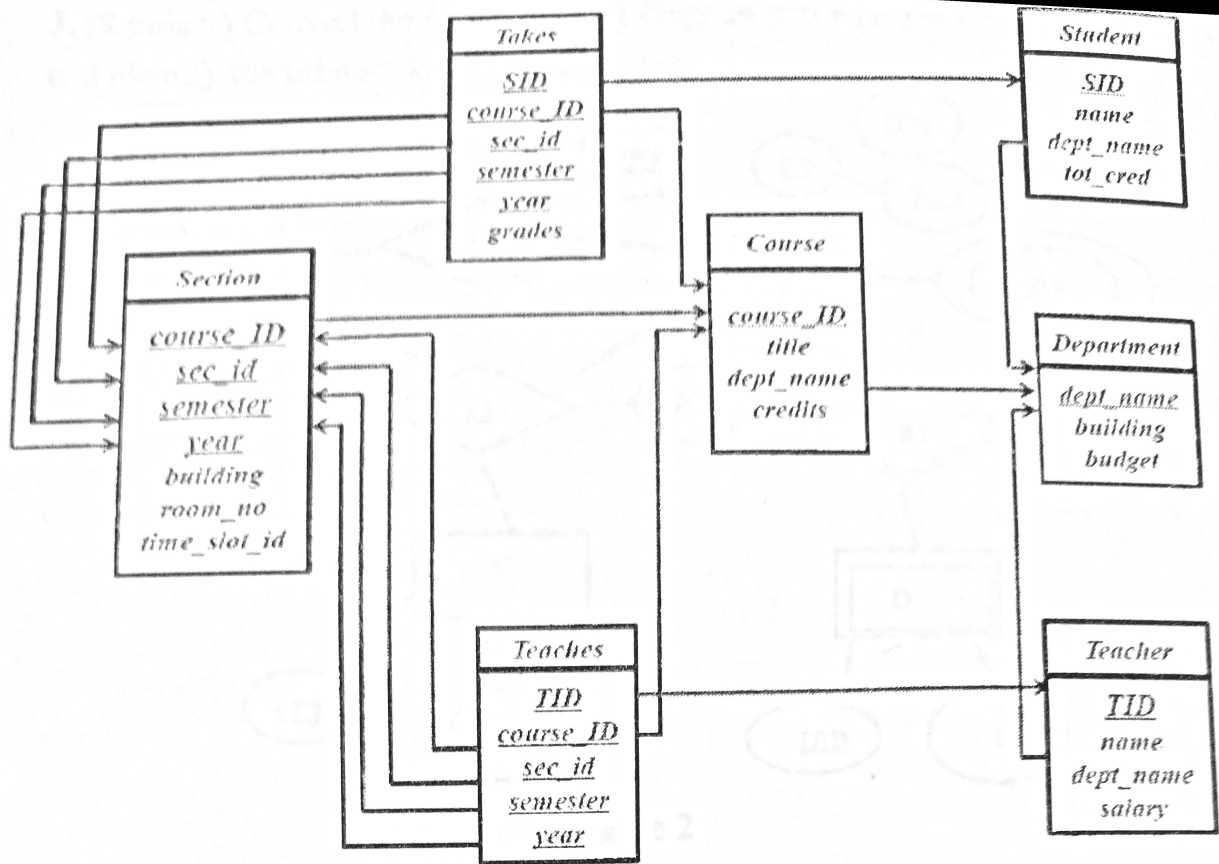


Figure 1

(1) Use a SQL statement to find the names of all students who are in CS department and have *tot_cred* values greater than that of some one student in EE department. (5 points)

if the total number of the courses he takes in the 2016 Spring semester is greater than 2, then give a SQL statement or SQL statements to increase the *tot_cred* (i.e. the total credits hours the student earned so far) by 10 percent. (6 points)

(3) Give a relational algebra expression, to find the *names* of all teachers belonging to the departments located in *Building3*. (3 points)

(4) To speed up the query in (3), use SQL statements to define some indexes on the tables involved in this query. (4 points)

Design the E-R Diagram on the basis of the information above.

3. (8 points) Convert the following E-R diagram to the proper relation schemas and identify the primary key of each relation.

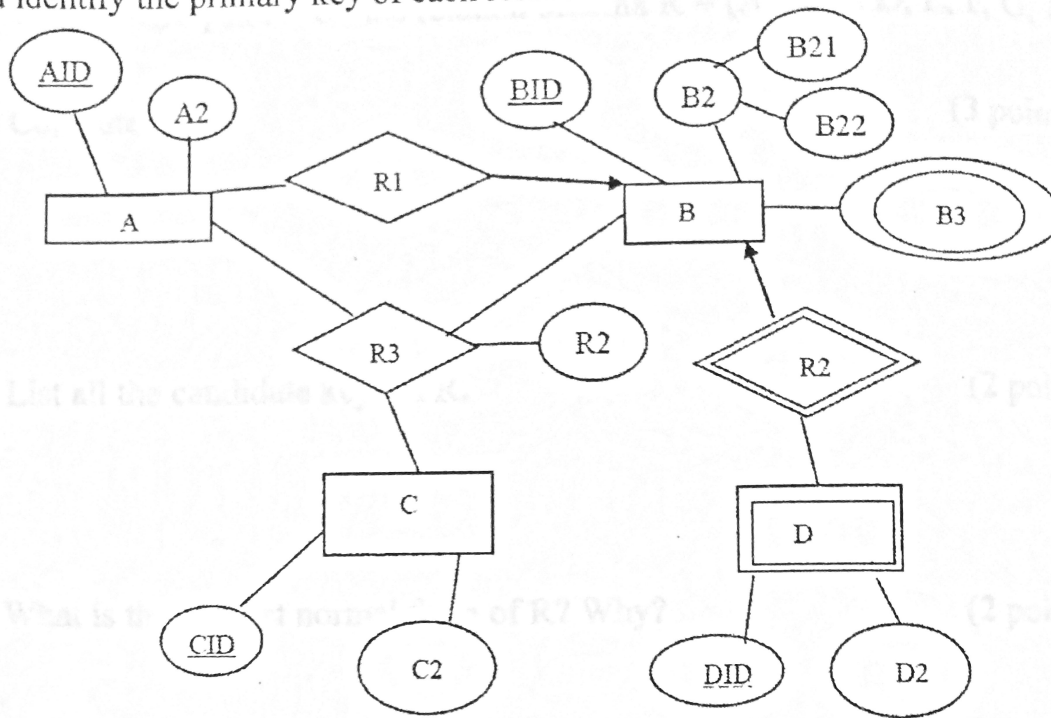


Figure 2

4. (10 points) Consider the following information about the pharmacies, patients and drugs:

- (1) Patients are identified by an SSN, and their names, addresses, and ages must be recorded.
- (2) Doctors are identified by an SSN. For each doctor, the name, specialty, and years of experience must be recorded.
- (3) Each pharmaceutical company (制药公司) is identified by name and has a phone number.
- (4) For each drug, the trade name and formula (成份) must be recorded. Each drug is produced by a given pharmaceutical company, and the trade name identifies a drug uniquely from among the products of that company.
- (5) Each pharmacy(药房) has a name, address, and phone number. Each pharmacy is identified by ID.
- (6) Every patient has a primary doctor. Every doctor has at least one patient.
- (7) Each pharmacy sells several drugs and has a price for each. A drug could be sold at several pharmacies, and the price could vary from one pharmacy to another.
- (8) Doctors prescribe drugs for patients. A doctor could prescribe one or more drugs for several patients, and a patient could obtain prescriptions from several doctors. Each prescription has a date and a quantity associated with it.
- (9) Pharmaceutical companies have long term contracts with pharmacies. A pharmaceutical company can contract with several pharmacies, and a pharmacy can contract with several pharmaceutical companies. For each contract, you have to store a start date, an end date.

Design the E-R Diagram on the basis of the information above.

5. (12 points) The functional dependency set $F = \{ AB \rightarrow C, A \rightarrow DEI, B \rightarrow FH, F \rightarrow GH, D \rightarrow IN \}$ holds on the relation schema $R = (A, B, C, D, E, F, G, H, I, N)$,
(3 points)

(1) Compute $(AD)^+$

(2) List all the candidate keys of R.

(2 points)

(3) What is the highest normal form of R? Why?

(2 points)

(4) Compute the canonical cover F_c .

(2 points)

(5) Give a lossless and dependency-preserving decomposition of R into 3NF.
(3 points)

6. (6 points) Give the data file `student(student_dept, student_ID, student_name)` as shown below, which is organized as a sequential file, taking the attribute `student_dept` as the search key.

(1) Is the index a dense or sparse index, why?

(3 points)

- (2) If a tuple (IS, 0430, Zhu) is deleted from the indexed file, depict the indexed file and the index file. (3 points)

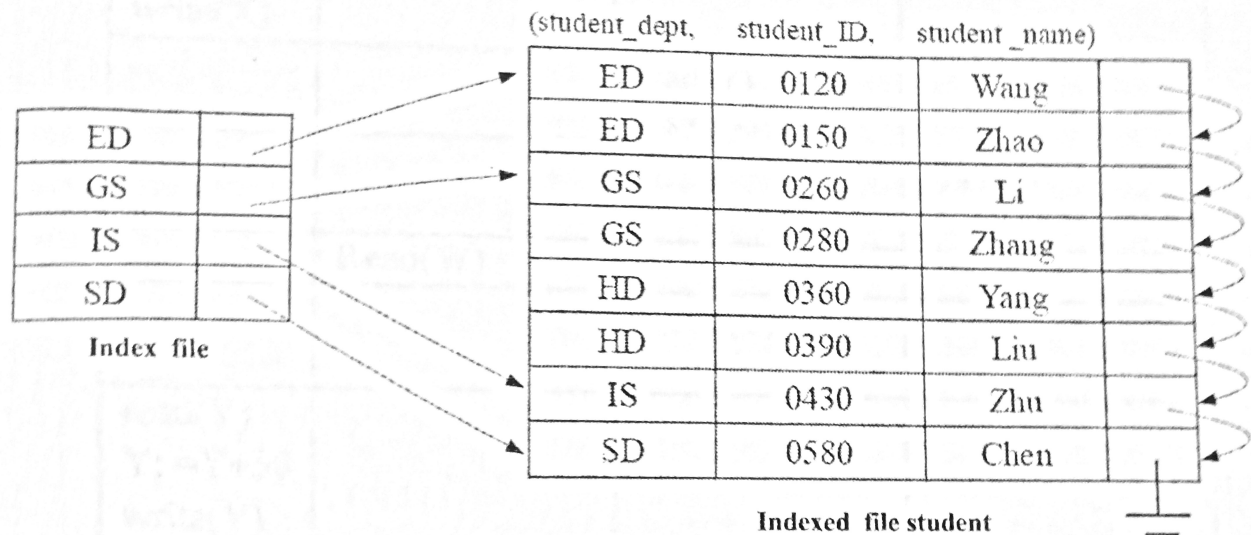


Figure 3

7. (10 points) Consider the database *University* given in Figure 1.

- (1) Give a SQL statement to find some students and list their names and departments that they belong to. It is required that their total credits (presented by *tot_cred*) are more than 40, their departments are located in *Building 3*, and they take the course identified by '2016CSDBS'. (3 points)

(2) For the SQL statement in (1), give an optimized query tree. (7 points)

(3) When T_1 is modifying *Course*, transaction T_2 wants to access *Course* by
 select *
 from *Course*
 where id in ('OS', 'DBS', 'Computer')
 does what type of three-level locks should T_2 request? Among these requests, which will be granted, and which will be refused? (3 points)

8. (4 points) Is the concurrent schedule S2 a recoverable schedule, and why?

S2:

T_1	T_2	T_3
read(A) $A := A - 1$ write(A)		
	read(B)	
commit		
		read(A) $A := A + 1$ write(A)
	$B := B - 1$ write(B)	
	read(A)	
		commit
	$A := A + 1$ write(A) commit	

9. (4 points) Is S3 a cascadeless schedule, and why?

S3:

T ₁	T ₂	T ₃	T ₄
read(X) X:=X-10 write(X)			
		read(Y) Y:=Y-10	
			read(Z) Z:=Z+10
	Read(W)		
		Write(Y) commit	
read(Y) Y:=Y+50 write(Y)			
	W:=W-20 write(W) commit		
commit			
			write(Z) read(W) W:=W+10 write(W) commit

10. (8 points) In the *University* database system, it is assumed that multiple granularity locks are granted on three-level data items, i.e. the database, tables, and tuples.

(4 points) If a transaction T₁ is now modifying the table *Course* by,

```
update Course
set credits=credits+1
where title='DBS'
```

, then what types of locks on the database, tables and tuples should it hold?

(2) When T_1 is modifying *Course*, transaction T_2 wants to access *Course* by
select *

from *Course*

where *title* in {'OS', 'DBS', 'Compiler'}

, then what types of three-level locks should T_2 request? Among these lock requests, which will be granted, and which will be refused? (5 points)

11. (10 points) Considering the concurrent schedule S4 for transactions T_1 , T_2 , T_3 and T_4 , and the data items A, B, C, D modified by these transactions. It is assumed that the initial values of these data items are A=4, B=0, C=30, D=10. When a failure occurs, the log-based recovery scheme consults the log to determine the recovery operations (i.e. redo, undo, ignore) done on T_1 , T_2 , T_3 and T_4 .

(1) Write the log file records for schedule S4. (4 points)

(2) After recovery operations on T_1 , T_2 , T_3 and T_4 are completed, what are the values of the data items A, B, C, D in the database? (4 points)

(3) Write down the redo, undo, ignore list of transactions respectively.(2 points)

S4:

T ₁	T ₂	T ₃	T ₄	recovery
begin transaction				
read(A) A:=A+6 write(A)				
	begin transaction read(D)			
commit				
	D:=D+10 write(D)			
		begin transaction read(B) B:=B+5 write(B)		
				checkpoint
			begin transaction read(C)	
	read(A) A:=A+10 write(A)			
	commit			
			C:=C+10 write(C)	
		read(A) A:=A+30 write(A)		
		commit		
			read(D) D:=D-10 write(D)	
*** Failure ***				